



NATIONAL UNIVERSITY OF MODERN LANGUAGES

SOFTWARE QUALITY ENGINEERING LAB FINAL

NAME: Abdul Rafay

ROLL NO: FL23791

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COURSE: SOFTWARE QUALITY ENGINEERING

SUBMITTED TO: Mam Rubab

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SOFTWARE QUALITY ASSURANCE LAB EXAM

Question 1: Black Box Testing (Boundary Value Analysis and Equivalence Class Partitioning)

Equivalence Class Partitioning (ECP)

Equivalence Class Partitioning is a black box testing technique in which input values are divided into valid and invalid groups called equivalence classes. One value from each class is selected for testing because all values in the same class are expected to behave similarly.

A. Student Age (18–30 Years)

Equivalence Class	Range	Type
EC1	Less than 18	Invalid
EC2	18 to 30	Valid
EC3	Greater than 30	Invalid

B. Number of Courses Selected (3–6)

Equivalence Class	Range	Type
EC4	Less than 3	Invalid
EC5	3 to 6	Valid
EC6	Greater than 6	Invalid

C. Semester Fee Amount (PKR 10,000–100,000)

Equivalence Class	Range	Type
EC7	Less than 10,000	Invalid
EC8	10,000 to 100,000	Valid
EC9	Greater than 100,000	Invalid

Boundary Value Analysis (BVA)

Boundary Value Analysis is used to test values at the lower boundary, upper boundary, and values just below and above the boundaries.

Student Age Test Cases

Test Case ID	Input Field	Test Value	Expected Result	Test Type	Status
TC-A1	Student Age	17	Age should be rejected	BVA	Invalid
TC-A2	Student Age	18	Age should be accepted	BVA	Valid
TC-A3	Student Age	19	Age should be accepted	BVA	Valid
TC-A4	Student Age	29	Age should be accepted	BVA	Valid
TC-A5	Student Age	30	Age should be accepted	BVA	Valid
TC-A6	Student Age	31	Age should be rejected	BVA	Invalid

Number of Courses Test Cases

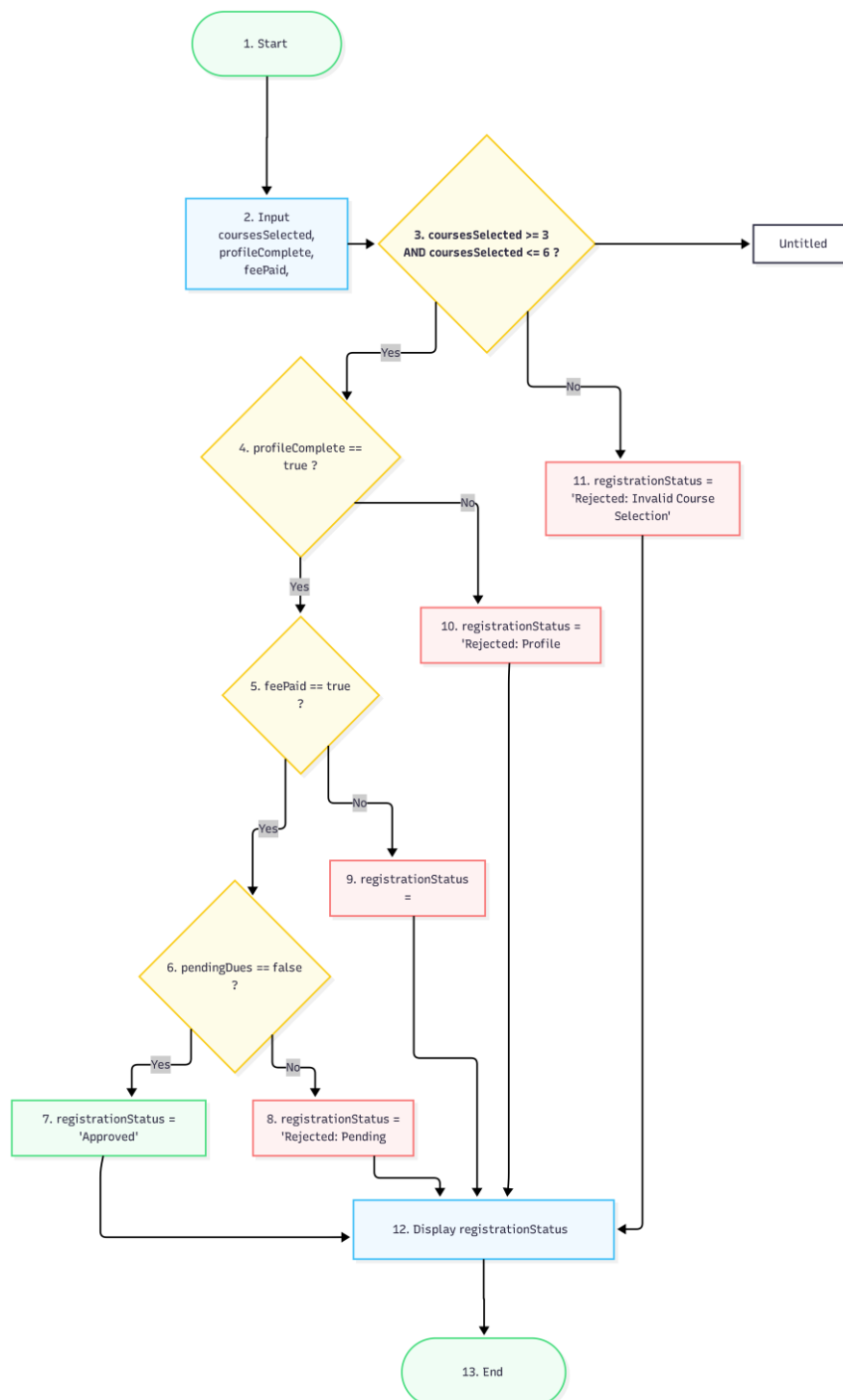
Test Case ID	Input Field	Test Value	Expected Result	Test Type	Status
TC-C1	Courses Selected	2	Selection should be rejected	BVA	Invalid
TC-C2	Courses Selected	3	Selection should be accepted	BVA	Valid
TC-C3	Courses Selected	4	Selection should be accepted	BVA	Valid
TC-C4	Courses Selected	5	Selection should be accepted	BVA	Valid
TC-C5	Courses Selected	6	Selection should be accepted	BVA	Valid
TC-C6	Courses Selected	7	Selection should be rejected	BVA	Invalid

Semester Fee Test Cases

Test Case ID	Input Field	Test Value	Expected Result	Test Type	Status
TC-F1	Semester Fee	9,999	Fee should be rejected	BVA	Invalid
TC-F2	Semester Fee	10,000	Fee should be accepted	BVA	Valid
TC-F3	Semester Fee	10,001	Fee should be accepted	BVA	Valid
TC-F4	Semester Fee	99,999	Fee should be accepted	BVA	Valid
TC-F5	Semester Fee	100,000	Fee should be accepted	BVA	Valid
TC-F6	Semester Fee	100,001	Fee should be rejected	BVA	Invalid

Question 2: White Box Testing

Control Flow Graph (CFG)



Total Decision Nodes = 4

Cyclomatic Complexity

Cyclomatic Complexity is used to determine the number of independent paths in a program.

Formula

$$V(G) = \text{Number of Decision Nodes} + 1$$

Calculation

$$V(G) = 4 + 1$$

Result

$$\text{Cyclomatic Complexity} = 5$$

Therefore, 5 independent paths are required for complete path coverage.

Independent Paths

Path 1 (Successful Registration)

- Valid course selected
- Profile completed
- Fee paid
- No pending dues
- Registration approved

Path 2 (Rejected Due to Pending Dues)

- Valid course selected
- Profile completed
- Fee paid
- Pending dues exist
- Registration rejected

Path 3 (Rejected Due to Unpaid Fee)

- Valid course selected
- Profile completed
- Fee not paid
- Registration rejected

Path 4 (Rejected Due to Incomplete Profile)

- Valid course selected
- Profile incomplete
- Registration rejected

Path 5 (Rejected Due to Invalid Course Selection)

- Invalid course selected
- Registration rejected

Path Coverage Test Cases

Test Case ID	Courses Selected	Profile Complete	Fee Paid	Pending Dues	Expected Result	Path
PTC-01	4	True	True	False	Registration Approved	P1
PTC-02	4	True	True	True	Rejected: Pending Dues	P2
PTC-03	4	True	False	False	Rejected: Fee Not Paid	P3
PTC-04	4	False	True	False	Rejected: Profile Incomplete	P4
PTC-05	2	True	True	False	Rejected: Invalid Course Selection	P5

Question 3: Test Plan for Student Course Registration System

Test Plan ID

TP-SCRS-001

Project Name

Student Course Registration System

Objective

The objective of this test plan is to verify that all functions of the Student Course Registration System work according to the specified requirements and provide correct results to users.

Scope

This testing covers login functionality, profile management, course registration, fee calculation, fee payment, and registration approval processes.

Features to be Tested

1. Student login with valid credentials.
2. Rejection of invalid login attempts.
3. Profile update functionality.
4. Student age validation.
5. Course selection validation.

6. Semester fee calculation.
7. Online fee payment.
8. Registration approval process.

Features Not to be Tested

1. Database performance testing.
2. Network performance testing.
3. Security penetration testing.
4. Third-party payment gateway internal operations.

Test Strategy

The following testing techniques will be used:

- Functional Testing
- Black Box Testing
- White Box Testing
- Boundary Value Analysis
- Equivalence Class Partitioning
- Path Coverage Testing

Entry Criteria

- System requirements are approved.
- Test environment is available.
- Software build is ready for testing.
- Test data has been prepared.

Exit Criteria

- All planned test cases have been executed.
- Critical defects have been fixed.
- Test results have been documented.
- Test summary report has been completed.

Test Deliverables

- Test Plan
- Test Cases
- Test Execution Report
- Defect Report

- Test Summary Report

Risks and Mitigation

Risk	Mitigation
Requirement changes during testing	Review requirements regularly
Test environment failure	Maintain backup environment
Lack of test data	Prepare test data before testing
Limited testing time	Prioritize critical functionalities

Requirement Traceability Matrix (RTM)

Requirement ID	Description	Test Case ID
FR-01	Valid Login	TC-01
FR-02	Invalid Login	TC-02
FR-03	Profile Update	TC-03
FR-04	Age Validation	TC-04
FR-05	Course Selection Validation	TC-05
FR-06	Fee Calculation	TC-06
FR-07	Online Fee Payment	TC-07
FR-08	Registration Approval	TC-08

Functional Test Cases

TC-01

Field	Description
Requirement ID	FR-01
Test Scenario	Login with valid credentials
Test Steps	Enter valid student ID and password and click Login
Test Data	ST123, Pass123
Expected Result	User should successfully log in

TC-02

Field	Description
Requirement ID	FR-02
Test Scenario	Login with invalid credentials
Test Steps	Enter incorrect password and click Login
Test Data	ST123, Wrong123
Expected Result	Login should be rejected

TC-03

Field	Description
Requirement ID	FR-03
Test Scenario	Update profile information
Test Steps	Edit profile information and save
Test Data	Updated profile details
Expected Result	Profile should be updated successfully

TC-04

Field	Description
Requirement ID	FR-04
Test Scenario	Age validation
Test Steps	Enter age less than 18
Test Data	17
Expected Result	Validation error message should appear

TC-05

Field	Description
Requirement ID	FR-05
Test Scenario	Invalid course selection
Test Steps	Select fewer than 3 courses
Test Data	2 courses
Expected Result	Registration should not proceed

TC-06

Field	Description
Requirement ID	FR-06
Test Scenario	Semester fee calculation
Test Steps	Select courses and calculate fee
Test Data	4 courses
Expected Result	Correct semester fee should be displayed

TC-07

Field	Description
Requirement ID	FR-07
Test Scenario	Online fee payment
Test Steps	Pay semester fee online
Test Data	PKR 50,000
Expected Result	Payment should be completed successfully

TC-08

Field	Description
Requirement ID	FR-08
Test Scenario	Registration approval
Test Steps	Complete profile, pay fee, and clear dues
Test Data	Valid student information
Expected Result	Registration should be approved